Emerging Technologies and IT Audit

Legal, Ethical and Professional Issues in IT Auditing

IT Audit and Control

Learning Outcomes

- Explain how emerging technologies change IT audit risks and controls.
- Identify key audit areas for cloud, AI, IoT and blockchain systems.
- Distinguish continuous auditing from continuous monitoring.
- Explain how data analytics supports risk-based IT audit.
- Discuss legal, ethical and professional responsibilities of IT auditors.
- Apply ISACA and ISO standards to IT audit engagements.

Why Emerging Technologies Matter to IT Auditors

- Technology changes the location, ownership and complexity of controls.
- Traditional periodic audits may miss rapidly changing risks.
- New technologies create both new risks and new audit evidence.
- Auditors increasingly assess governance, algorithms, data, third parties and automated controls.
- Audit focus: Are systems secure, reliable, compliant, resilient and fit for purpose?

Emerging Technology Audit Landscape

Technology

Cloud: shared infrastructure and outsourced services

AI: automated decisions and prediction

IoT: connected devices and sensors

Blockchain: distributed, tamper-evident records

Typical Audit Questions

- Who owns the risk and control?
- Where is the data stored and processed?
- Can controls be independently tested?
- What happens when technology fails?
- Is there adequate audit evidence?

Auditing Cloud Computing

Key risks: data exposure, misconfiguration, weak identity management, vendor dependency and service outages.

Shared-responsibility model: determine which controls are operated by the provider and which remain with the customer.

Audit areas: Identity and Access Management (IAM), encryption, logging, backup/recovery, configuration management, change management and incident response.

Auditing Cloud Computing

Third-party assurance: review contracts, service-level agreements and independent assurance reports.

Evidence: configuration records, access logs, provider reports, tickets, policies and recovery-test results.

Cloud Audit: Practical Checklist

Cloud Audit Checklist

- ✓ Asset and data inventory
- ✓ Identity and privileged access
- ✓ Security configurations
- ✓ Encryption and key management
- ✓ Logging and monitoring
- ✓ Backup and disaster recovery

Questions for Auditors

- ➢ Are cloud resources securely configured?
- ➢ Are inactive accounts removed promptly?
- ➢ Can management detect unusual activity?
- ➢ Are recovery objectives tested?
- ➢ Does the contract define audit rights?

Auditing Artificial Intelligence (AI) Systems

Key risks: biased outcomes, inaccurate outputs, opaque decisions, privacy violations, model drift and unauthorized use.

Governance: approved use cases, accountability, risk classification, human oversight and documented model ownership.

Data: quality, representativeness, lawful use and protection of sensitive information.

Model controls: testing, validation, monitoring, version control, change approval and performance thresholds.

Generative AI: assess confidentiality, hallucination risk, prompt/data leakage, intellectual property and human review.

Example

- Suppose a university uses an **AI** system to predict students who may fail a course. The auditor could test whether:
  - The model was validated before deployment.
  - The training data was accurate and representative.
  - The model does not unfairly disadvantage certain students.
  - Changes to the model require authorisation.
  - Model performance is regularly monitored.
  - Lecturers can review and override AI recommendations.
  - All model changes and important decisions are logged.

Class Question: Auditing an AI System

A university has introduced an AI system that predicts which students are at risk of failing their courses.

The system analyses students' attendance, assessment marks, LMS activity and previous academic performance.

Task:

As an IT auditor, identify five model controls that should be in place for this AI system.

For each control:

1.State the model control.

2.Explain why it is necessary.

3.Give an example of audit evidence that you would examine.

AI Audit: What to Examine

AI Audit Domains

- Governance and accountability
- Data quality and lineage
- Fairness and bias
- Security and privacy
- Model performance
- Explainability and transparency

Evidence

- Model documentation
- Training/validation data records
- Test results and validation reports
- Access and usage logs
- Change/version history
- Human-oversight records

Auditing Internet of Things (IoT)

- IoT combines devices, networks, applications, cloud services and physical processes.
- Key risks: insecure devices, default credentials, unpatched firmware, weak communications and physical tampering.
- Audit the complete ecosystem—not only the device.
- Check device inventory, authentication, network segmentation, firmware updates, encryption and monitoring.
- Consider safety and operational impacts where IoT controls physical equipment.

Auditing Blockchain Systems

- Blockchain provides a distributed record, but it does not automatically make data correct or systems secure.
- Audit governance: who can participate, validate transactions or change rules?
- Assess consensus mechanisms, smart contracts, private keys, wallet controls and access management.

Auditing Blockchain Systems

- Check the accuracy of data entering the blockchain—the 'oracle/input' problem.
- Audit smart contracts for logic errors, unauthorized functions and inadequate testing.
- Consider privacy, legal enforceability, interoperability and recovery arrangements.

Comparing Emerging Technology Audit Risks

Emerging Technology

- Cloud → third-party and configuration risk
- AI → model/data and decision risk
- IoT → device and physical risk
- Blockchain → governance and smart-contract risk

Common Audit Response

- Identify assets and owners
- Map risks to controls
- Test preventive and detective controls
- Validate evidence and logs
- Assess resilience and compliance
- Report residual risk

Continuous Auditing and Continuous Monitoring

- Continuous auditing uses technology and analytics to perform audit procedures more frequently or continuously.
- Continuous monitoring is management's ongoing process for observing controls, risks and performance.
- Continuous audit supports independent assurance; continuous monitoring is primarily a management responsibility.

Examples: automated access reviews, exception alerts, configuration checks and transaction analytics.

Goal: move from periodic detection toward earlier identification and response.

Continuous Audit Cycle

1. Define risk indicators and control objectives.
2. Connect to relevant systems and data sources.
3. Collect data continuously or at high frequency.
4. Apply rules, thresholds and analytics to identify exceptions.
5. Investigate and validate significant exceptions.
6. Report findings, assign actions and track remediation.
7. Refine tests as risks and systems change.

Use of Data Analytics in IT Audit

- Data analytics allows auditors to examine large volumes of transactions, logs and configuration data.
  - **Descriptive**: What happened?
  - **Diagnostic:** Why did it happen?
  - **Predictive:** What may happen?
  - **Prescriptive**: What action could reduce the risk?

Examples: duplicate payments, unusual logins, inactive accounts, privilege anomalies, failed controls and configuration deviations.

Data Analytics Audit Workflow

Plan → define the audit objective and risk.

Acquire → identify and obtain relevant data.

Prepare → clean, validate and document data quality.

Analyse → use queries, statistics, rules, visualisation or machine learning.

Investigate → determine whether exceptions represent actual control failures.

Conclude → document evidence, limitations and audit findings.

Protect → secure audit data and maintain confidentiality.

Question

Explain the importance of data analytics in IT auditing. In your answer:

1. Identify three benefits of using data analytics in IT audit.
2. Give two examples of how data analytics can help an IT auditor identify risks or control weaknesses.
3. Explain one limitation or challenge of using data analytics in IT audit.

Legal, Ethical and Professional Issues

Responsibilities of the IT Auditor in a Digital Environment

IT Audit and Control

IT Laws and Regulations

IT auditors must understand the laws and regulations that apply to the organisation, sector and jurisdiction.

Data protection requirements commonly address lawful processing, purpose limitation, security, retention, data subject rights and breach management.

Other relevant areas may include cybercrime, electronic transactions, intellectual property, records retention and sector-specific regulation.

Audit question: Are legal obligations translated into policies, procedures and technical controls?

For Zambia, consider the Data Protection Act, 2021 and other applicable ICT/cyber laws and regulations.

Data Protection: What Should Be Audited?

Data Protection Audit Areas

- Data inventory and classification
- Lawful basis /authorised use
- Access controls
- Encryption and security
- Retention and disposal
- Third-party processing

Audit Evidence

- Privacy policies
- Processing records
- Access logs
- Contracts and data-processing clauses
- Incident/breach records
- Data retention schedules

Professional Ethics in IT Auditing

- **Integrity** — be honest, objective and truthful in audit work.
- **Objectivity** — avoid bias, conflicts of interest and undue influence.
- **Confidentiality** — protect information obtained during the engagement.
- **Professional competence** — maintain the knowledge and skills needed for the assignment.
- **Professional care** — plan, perform and document work carefully.
- **Professional behaviour** — comply with applicable laws, standards and professional requirements.

Auditor Independence and Responsibility

Independence supports credible and unbiased audit conclusions.

- Avoid auditing systems or controls where personal, financial or management interests create conflicts.
- Disclose conflicts of interest before or during the engagement.
- Do not allow management pressure to determine audit findings or conclusions.
- Maintain sufficient professional scepticism: verify evidence rather than accepting management assertions.
- Responsibility includes protecting evidence, documenting judgments and communicating significant risks.

ISACA Standards and Frameworks

- ISACA's IT Audit Framework (ITAF), 5th Edition, provides a professional practices framework for IT audit and assurance.
- ITAF addresses roles and responsibilities, ethics, professional behaviour, knowledge and skills, and guidance for planning, performing and reporting engagements.
- COBIT provides a governance and management framework for enterprise information and technology.
- ISACA resources increasingly address cloud, AI, data analytics, continuous assurance and digital trust.
- Use standards and guidance together with organisational policies and applicable laws.

Relevant ISO Standards

- **ISO 19011:2026** — guidelines for auditing management systems, including audit principles, audit programmes, competence and conducting audits.
- **ISO/IEC 27001** — requirements for an Information Security Management System (ISMS).
- **ISO/IEC 27002** — information security controls and implementation guidance.
- **ISO/IEC 27007** — guidance for auditing an ISMS.

Select standards according to the audit objective; do not treat every ISO standard as a mandatory legal requirement.

Good Audit Practice

- **Risk-based** — focus effort on areas of greatest risk and impact.
- **Evidence-based** — conclusions must be supported by sufficient and appropriate evidence.
- I**ndependent** — maintain impartiality and manage conflicts.
- **Secure** — protect audit data, credentials and working papers.
- **Traceable** — document procedures, evidence, judgments and conclusions.
- **Technology-aware** — understand how emerging technologies affect controls and audit evidence.
- **Action-oriented** — communicate root causes, risks and practical recommendations.

Class Activity: Audit an Emerging Technology

ACTIVITY

Choose one: cloud, AI, IoT or blockchain.

1. Identify five major risks associated with the technology.
2. For each risk, propose one preventive and one detective control.
3. Identify evidence an auditor could collect to test each control.
4. Identify one relevant legal/regulatory requirement and one professional/technical standard.
5. Present a short audit plan: objective, scope, risks, controls, evidence and expected findings.

Key Takeaways

- Emerging technologies change both the risks being audited and the methods used to audit them.
- Cloud, AI, IoT and blockchain require technology-specific audit procedures.
- Continuous auditing and analytics can provide earlier and broader assurance.
- Legal compliance and professional ethics are integral to IT audit quality.
- Independence, competence, confidentiality and professional scepticism remain fundamental.
- ISACA and ISO standards provide structured guidance—but auditors must also consider applicable laws and organisational requirements.

References and Further Reading

ISACA (2026). IT Audit Framework (ITAF), 5th Edition.

ISACA (2026). IT Audit Resources and guidance on emerging technologies.

ISO (2026). ISO 19011:2026 — Guidelines for auditing management systems.

ISO/IEC 27001 — Information security management systems requirements.

ISO/IEC 27002 — Information security controls.

ISO/IEC 27007:2020 — Guidelines for information security management systems auditing.

Republic of Zambia. Data Protection Act, 2021.

Use the latest official editions and applicable national legislation when conducting an actual audit.

ISACA: https://www.isaca.org/resources/it-audit
ISO: https://www.iso.org/standard/88984.html